

3-1 Work and Power



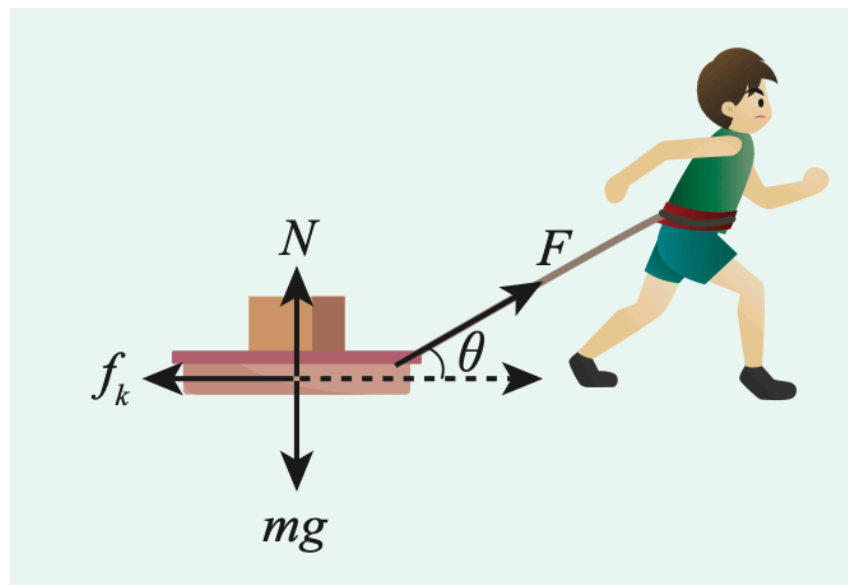
Source / 來源

- 選修物理(二) 課堂講義 (4624R51) 教用版，第3章 功與能量，§ 3-1 功與功率，pp. 104–115
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 7: Kinetic Energy and Work
- Young & Freedman, *University Physics with Modern Physics*, Ch 6: Work and Kinetic Energy

Examples

Example 1 — Work by a Constant Force (*p. 105*)

A person pulls a basin (mass 5 kg) along a horizontal surface using a rope at angle $\theta = 37^\circ$ above the horizontal. The applied force is $F = 50$ N, and the coefficient of kinetic friction between the basin and the surface is $\mu_k = 0.3$. Take $g = 10$ m/s².



When the basin is displaced 20 m horizontally, find:

- The work done by the rope on the basin.
- The work done by friction on the basin.

$$W_F = Fd \cos \theta = 50 \times 20 \times \cos 37^\circ = 50 \times 20 \times 0.8 = 800 \text{ J}$$

(c) The work done by the normal force on the basin.

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 Solution

Try It 1 (*p. 105*)

Following Example 1, when the basin has been displaced 1 m:

(a) Does the basin's energy increase, decrease, or remain unchanged?

(b) What is the net work done on the basin?

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 Solution

Example 2 — Net Work with Newton's Second Law (*p. 106*)

A block of mass M is suspended by a rope and lowered vertically with a downward acceleration of $\frac{g}{4}$ over a distance l . Find:

- (a) The work done by the rope (tension) on the block.
- (b) The work done by gravity on the block.
- (c) The net work done on the block.
- (d) Can we obtain the net work by first finding the net force via Newton's second law?

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 Solution

Try It 2 (p. 106)

A block of mass m is placed on a rough inclined plane (inclination angle θ , length L). The block slides down at **constant velocity** under gravity mg , normal force N , and kinetic friction f_k .

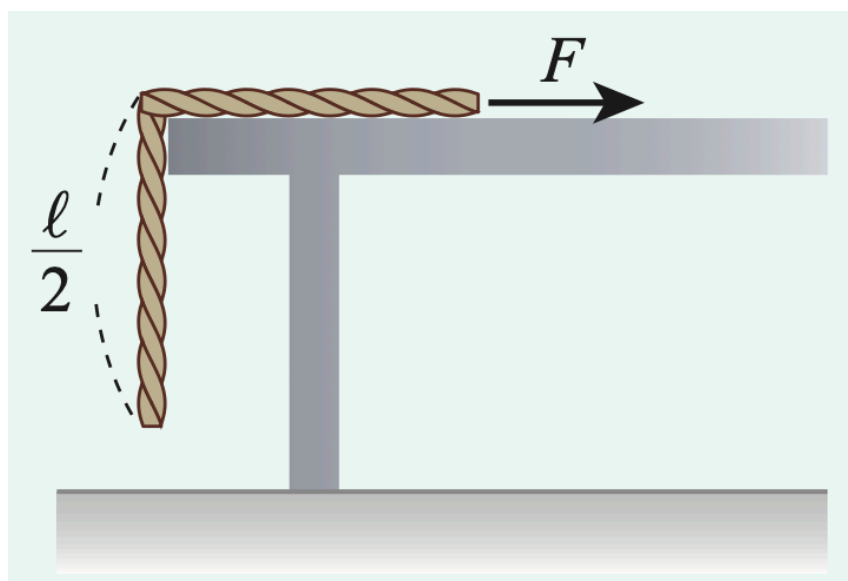
Find the work done by each force as the block slides from top to bottom:

- (a) Gravity
- (b) Normal force
- (c) Kinetic friction
- (d) Net force

💡 Solution

Example 3 — Work by a Variable Force (*p. 108*)

A uniform rope of mass m and length l lies on a frictionless horizontal table with half its length hanging over the edge. What is the minimum work required to pull the rope entirely onto the table?



💡 Solution

Try It 3 (p. 108)

A spring is slowly compressed by a distance x , requiring a force F and doing work W . If the spring is then compressed by an **additional** distance x (total compression = $2x$), which of the following are correct? (Select 3.)

- (A) The final force required is F .
- (B) The final force required is $2F$.
- (C) The additional work done is $2W$.
- (D) The additional work done is $3W$.
- (E) The additional work done is $\frac{3}{2}Fx$.

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 Solution

Example 4 — Instantaneous Power (p. 109)

A boat experiences water resistance proportional to its speed. At a constant speed of 10 m/s, the engine power is 20 hp (1 hp = 746 W).

- (a) Find the magnitude of the water resistance at this speed.
- (b) What engine power is needed to double the boat's speed?

 Solution

Try It 4 (p. 109)

An airplane in level flight experiences air drag $f = bv^2$ ($b > 0$). At constant speed v_0 , the engine balances the drag.

What is the ratio of the engine output power at speed $2v_0$ to that at speed v_0 ?

(Adapted from 96 指考)

 Solution

Quick Check

Work by a Constant Force

Q1. (p. 110) A block sits on a smooth inclined plane (angle θ). A horizontal force F is applied to the **incline**, causing the entire system to accelerate at a to the right. The block remains stationary relative to the incline.

Which of the following forces do work on the **block**? (*Select 2.*)

(A) Horizontal force F

(B) Gravity W

(C) Normal force N

(D) Friction f

(E) None of the above

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 Solution

Q2. (*p. 110*) A person pulls a sled (mass $m = 10$ kg) at constant velocity along a horizontal surface with a rope at angle $\theta = 45^\circ$ above horizontal. The coefficient of kinetic friction is $\mu = 0.2$ and $g = 10$ m/s². How much work does the person do over a distance of 30 m?

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 Solution

Q3. (*p. 110*) An object slides down an inclined plane. From a ground-frame observer, does the contact force from the incline do work on the object? Fill in "Yes" or "No":

Condition	Work done?
(a) Smooth incline, fixed to the ground.	
(b) Smooth incline, free to slide on a frictionless floor.	
(c) Rough incline, fixed to the ground.	
(d) Rough incline, free to slide on a frictionless floor.	

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 Solution

Work by a Variable Force

Q4. (*p. 111*) A 2 kg object, initially at rest, is acted upon by an external force. The F - x graph shows $F = 20$ N at $x = 0$, linearly decreasing to $F = 0$ at $x = 10$ m. What is the magnitude of work done by the external force from $x = 0$ to $x = 10$ m?

(A) 20 J

(B) 40 J

(C) 100 J

(D) 120 J

(E) 200 J

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 Solution

Power

Q5. (*p. 111*) An electric motor drives an elevator. When the cable tension is 1800 N, the elevator ascends at a constant speed of 0.5 m/s.

(a) What is the work done on the elevator in 10 s?

(b) What is the average power?

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 Solution

Q6. (*p. 111*) A person rides a bicycle on flat ground at a constant speed of 5 m/s with a pedaling power of 80 W. What is the resistance force from the ground?

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 Solution

Q7. (*p. 111*) Following Q6: the person and bicycle have a total mass of 60 kg. When riding up a slope with $\sin \theta = 0.02$, the friction resistance is unchanged and the pedaling power remains 80 W. What is the constant speed going uphill? ($g = 10 \text{ m/s}^2$)

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 Solution

Challenge

Work by a Constant Force

Q1. (*p. 112*) Three blocks A, B, C with masses m , $2m$, $3m$ are on a horizontal surface (kinetic friction coefficient $\mu = 0.5$). A horizontal force $8mg$ is applied to block A, pushing the entire group a distance L .

Which of the following are correct? (*Select 3.*)

(A) The work done by A on B is $\frac{20}{3}mgL$.

(B) The work done by B on C is $2mgL$.

(C) The work done by the horizontal force on A is $8mgL$.

(D) The work done by friction on A is $-\frac{1}{2}mgL$.

(E) The net work on the entire group is $3mgL$.

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 Solution

Q2. (*p. 112*) A 10 kg object starts from rest. A force of 50 N is applied at 37° above the horizontal for 2.0 s. Given $g = 10 \text{ m/s}^2$, $\mu_k = 0.2$, find:

(a) The work done by the applied force.

(b) The work done by friction.

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 Solution

Q3. (*p. 112*) A 4 kg mass on a smooth horizontal surface is connected by a massless rope over a frictionless pulley. A constant downward force of 10 N is applied at the other end. The mass slides from A to B along the surface.

[Figure: Pulley at top. Rope makes 37° with horizontal at A and 30° with vertical at the pulley. Horizontal distance A to B is 2 m.]

Find the work done by the applied force.

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 Solution

Work by a Variable Force (*pp. 112–113, 題組*)

Q4. A satellite orbits Earth in a circular orbit. What is the work done by the gravitational force on the satellite over one complete orbit?

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 Solution

Q5. A person holds a heavy object stationary in the air. What is the work done by the person's hand?

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 Solution

Q6. On a frictionless horizontal surface, an object undergoes uniform circular motion. What is the work done by the centripetal force?

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 Solution

Q7. On a sandy horizontal surface (coefficient of kinetic friction μ), a force maintains an object of mass m in uniform circular motion of radius r for one full revolution. Find the work done by the applied force.

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 Solution

Power

Q8. (*p. 113*) A mass of 4 kg is projected horizontally with initial speed 5 m/s from a certain height. Take $g = 10 \text{ m/s}^2$. When the velocity angle below horizontal reaches 37° , let P_1 be the **average** power of gravity from launch to that moment, and P_2 be the **instantaneous** power of gravity at that moment. Find P_1/P_2 .

(A) 1

(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{4}$

(E) 4

 Solution

Q9. (*p. 113*) An airplane in long-range level flight has lift $F_{\text{lift}} = \alpha \rho v^2$ and drag $F_{\text{drag}} = \beta \rho v$, where ρ is air density, v is airspeed, and α, β are constants. At high altitude, $\rho_H = \frac{1}{2} \rho_L$. If the airplane flies the same horizontal distance at each altitude, what is the ratio of energy consumed by drag (high : low)?

(A) $\frac{1}{2\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2}$

(D) 1

(E) $\sqrt{2}$

 Solution

Q10. (*pp. 113–114*) A car drives up a slope (angle $\theta = 30^\circ$) at constant speed v . At this speed, the drag force equals 0.5 times the car's weight, and drag is proportional to speed. If the car drives **down** the same slope at the same engine power, what is the maximum speed?

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 Solution

Exam Archive

Q1 — Governor Power Dissipation (*p. 114, 110 指考*)

A speed governor has a fixed cylindrical shell and a central rotating shaft. Two massless rods (length 0.40 m) extend from the shaft, each holding a 1.0 kg bob. When the half-angle is 45° , the bobs touch the inner wall (inner radius = 0.40 m). Given $g = 10 \text{ m/s}^2$ and $\mu = 0.25$ between bob and shell, find the power dissipated by friction per bob when $\omega = 6 \text{ rad/s}$.

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Q2 — Average Power of a Variable Force (*p. 114, 109 指考補考*)

A 1 kg object rests on the ground. A vertically upward force F varies linearly with time: $F = 9$ N at $t = 2$ s, increasing to $F = 18$ N at $t = 10$ s (and $F = 0$ for $t < 2$ s).

Find the average power of F over $t = 0$ to 10 s.

($g = 10 \text{ m/s}^2$; ignore air resistance.)

(A) 162 W

(B) 144 W

(C) 94.5 W

(D) 32.4 W

(E) 28.8 W

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Q3 — Pendulum Work (p. 115, 98 指考, 答對率 64%)

A pendulum swings back and forth. The bob experiences gravity W , string tension T , and air resistance R . Which statement is correct?

- (A) W always does positive work; T always does negative work.
- (B) R always does negative work; T always does no work.
- (C) W always does positive work; R always does negative work.
- (D) T and W always do positive work; R always does negative work.
- (E) W and R always do negative work; T always does no work.

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 Solution

Q4 — Airplane Drag Analysis (p. 115, 96 指考, 全對率 25%)

An airplane in level flight experiences drag $f = -bv$ ($b > 0$). Which of the following are correct? (*Multiple select.*)

- (A) At constant speed, the engine thrust equals the drag in magnitude.
- (B) At constant speed $2v_0$, the required thrust is twice that at v_0 .
- (C) At constant speed in level flight, the lift equals the weight.

(D) The power dissipated by drag is independent of speed.

(E) At constant speed $2v_0$, the engine output power is twice that at v_0 .

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 Solution

Glossary

English	中文	Symbol / Unit
work	功	W [J]
displacement	位移	$d, \mathbf{d}$
force	力	$F, \mathbf{F}$ [N]
dot product (scalar product)	內積 (純量積)	$\mathbf{A} \cdot \mathbf{B}$
net work / total work	合力作功 / 總功	W_{net}
kinetic friction	動摩擦力	f_k
coefficient of kinetic friction	動摩擦係數	μ_k
normal force	正向力	N
gravity / gravitational force	重力	mg
tension	繩張力	T
variable force	變力	$F(x)$
spring constant	彈力 (力) 常數	k [N/m]
Hooke's law	虎克定律	$F = kx$
elastic potential energy (spring PE)	彈性能 (彈力位能)	$U_s = \frac{1}{2}kx^2$
power	功率	P [W]
instantaneous power	瞬時功率	$P = \mathbf{F} \cdot \mathbf{v}$
average power	平均功率	$\bar{P} = W / \Delta t$

joule	焦耳	$J = N \cdot m$
watt	瓦特	$W = J/s$
horsepower	馬力	hp (1 hp = 746 W)
centripetal force	向心力	$F_c = m\omega^2 r$
drag / air resistance	阻力 / 空氣阻力	f
inclined plane	斜面	—
pulley	滑輪	
Newton's second law	牛頓第二運動定律	$\sum F = m\vec{a}$
lift (aerodynamic)	升力	F_{lift}
uniform circular motion	等速圓周運動	—